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MLS 101: Introduction to Medical Laboratory Science - Specimens, Safety and Accuracy

How laboratory results are produced, and why pre-analytical errors matter

FACULTY PRIMER

Faculty Primer

Faculty of Medical Laboratory Science
Medical Laboratory Science | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Most laboratory errors happen before the sample reaches the analyser. This pack introduces the laboratory disciplines, specimen collection and handling, biosafety, quality control and the meaning of reference ranges, with practical checklists for the teaching laboratory.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the major laboratory disciplines and the specimen types each handles.
- Explain the pre-analytical, analytical and post-analytical phases of testing.
- Apply biosafety and waste-disposal rules in a teaching laboratory.
- Interpret quality-control results and explain reference ranges.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The laboratory disciplines

Medical laboratory science is a set of specialised diagnostic disciplines, each with its own methods and quality concerns.

- Haematology: blood counts, haemoglobin, coagulation and blood film morphology.
- Clinical chemistry: electrolytes, enzymes, glucose, renal and liver function.
- Medical microbiology: culture, identification and antimicrobial susceptibility testing.
- Histopathology and cytopathology: tissue and cell examination.
- Immunology and serology: antibody and antigen detection.

- Blood transfusion science: grouping, cross-matching and component therapy.

The three phases of testing

A result is only as reliable as the weakest phase in the chain.

- Pre-analytical: patient preparation, collection timing, correct tube, adequate volume, labelling, transport and storage.
- Analytical: calibration, reagent quality, instrument maintenance and technician competence.
- Post-analytical: result validation, reporting, critical-value communication and record retention.
- Haemolysis, delayed transport and under-filled tubes are common causes of rejected samples.
- Label at the bedside, never in advance, and never relabel a specimen.
- Critical results must be phoned to the clinician and documented.

> **Rejection is a safety feature**
> Rejecting a wrongly labelled sample protects the patient from a result belonging to someone else. Never 'fix' a label.

Biosafety and quality control

The laboratory is a controlled environment with rules that exist for a reason.

- Treat all specimens as potentially infectious.
- No eating, drinking, applying cosmetics or storing food in the laboratory.
- Use a biological safety cabinet for procedures that generate aerosols.
- Decontaminate surfaces before and after work; separate clean from dirty zones.
- Sharps into puncture-resistant containers; chemical and biological waste segregated.
- Quality control: run control materials, plot Levey-Jennings charts and stop testing when results fall outside acceptable limits.

The laboratory's place in diagnosis

Laboratory results influence most clinical decisions, which makes accuracy a patient safety issue. The discipline covers haematology, clinical chemistry, microbiology, histopathology, immunology and blood transfusion, each with its own methods and quality requirements.

Every result passes through a chain: patient identification, collection, transport, preparation, analysis, validation and reporting. Most errors occur before the analyser is involved - wrong tube, insufficient volume, delayed transport, mislabelling.

Reference ranges describe the central 95 per cent of a healthy population; they are not thresholds of health. A result outside the range may be normal for that individual, and interpretation always requires clinical context.

Specimen quality and safety

Collection technique determines result quality. Venepuncture requires the correct order of draw to avoid cross-contamination between tube additives, adequate mixing, and labelling at the bedside. Haemolysed samples distort potassium and other analytes.

Urine, stool, sputum and swab samples each have collection requirements, and a poor sample produces a misleading result regardless of analyser quality.

Biosafety is daily discipline: gloves, lab coats, eye protection, no eating or drinking, correct decontamination, and safe disposal. Universal precautions assume every sample is infectious.

> **Label at the bedside**
> A specimen labelled away from the patient may belong to someone else. No result is worth that risk.

Quality control and assurance

Quality control runs samples of known value alongside patient samples. Control results plotted on a chart reveal drift and shifts before they affect patients. A failed control means results are held until the cause is found.

Quality assurance is broader: staff competence, equipment maintenance, calibration, documentation and external proficiency testing, where the laboratory is assessed against peers on the same samples.

Accreditation is external recognition that a laboratory meets a defined standard. It requires documented procedures, records, audits and corrective action - not good intentions.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Investigating a discrepant result

A patient's potassium is markedly raised but the patient has no symptoms.

1. Check for haemolysis - the commonest cause of spurious hyperkalaemia.
2. Check collection: tourniquet time, fist clenching, needle size.
3. Check for EDTA contamination from an earlier draw.
4. Check the delay between collection and analysis.
5. Request a repeat sample with correct technique.
6. Never release an implausible critical result without investigating.

> **What to carry forward**
> An implausible result is a specimen question before it is a patient question.

2. Reading a control chart

Controls drift upward for a week but stay within limits.

1. Recognise a trend rather than random variation.
2. Consider reagent ageing, lamp deterioration, calibration drift, temperature.
3. Check maintenance and calibration records.
4. Recalibrate and re-run controls.
5. Document the investigation and corrective action.
6. Review whether patient results in that period need retesting.

> **What to carry forward**
> Trends are warnings. Waiting for a control to fail means waiting for a wrong result.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain the pre-analytical, analytical and post-analytical phases and where errors occur.

20 marks

1. Define each phase with examples.
2. State that most errors are pre-analytical.
3. Give three pre-analytical errors and their effects.
4. Name controls that reduce them.
5. Conclude that quality is a pathway property.

Q2. Distinguish quality control from quality assurance.

10 marks

1. Control monitors method performance on known samples.
2. Assurance covers the wider system of competence and audit.
3. Explain why control alone is insufficient.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Reference range. The interval containing the central 95 per cent of healthy values.

Haemolysis. Red cell breakdown releasing contents that distort results.

Order of draw. The sequence for filling tubes to prevent additive cross-contamination.

Quality control. Running known samples to verify method performance.

Quality assurance. The whole system ensuring reliability from competence to audit.

Proficiency testing. External comparison of results with peer laboratories.

Calibration. Adjusting an instrument against known standards.

Critical value. A result requiring immediate clinical notification.

Asepsis. The absence of pathogenic microorganisms.

Universal precautions. Treating all samples as potentially infectious.

Accreditation. External recognition that a laboratory meets a standard.

Sensitivity. A test's ability to detect disease when present.

Your four-week plan

Week 1 - Pathway

Phases.

- Map the specimen pathway.
- List three pre-analytical errors.
- Explain reference ranges.

Week 2 - Collection

Technique.

- Practise order of draw.
- Label at the bedside.
- Identify haemolysis causes.

Week 3 - Safety

Biosafety.

- Demonstrate PPE use.
- Describe sharps disposal.
- Explain universal precautions.

Week 4 - Quality

Control.

- Interpret a control chart.
- Describe a corrective action.
- Explain accreditation.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any medical laboratory science text on collection and quality management.
- A clinical chemistry handbook on interferences and critical values.
- Your laboratory's biosafety manual.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Name the three phases of laboratory testing.

Answer: Pre-analytical, analytical and post-analytical.

2. Give three causes of specimen rejection.

Answer: Mislabelled specimen, inadequate volume, haemolysed sample, wrong container or delayed transport.

3. What is a reference range?

Answer: The interval within which results from a defined healthy population usually fall, used as a guide for interpretation and not a diagnosis.

4. Why are control materials run with patient samples?

Answer: To verify that the analytical system is performing within acceptable limits before patient results are reported.

5. State two biosafety rules for the laboratory.

Answer: Treat all specimens as infectious and never eat, drink or store food in the laboratory. Others include using PPE and decontaminating surfaces.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Match each laboratory discipline to two tests it performs.
- [ ] Write the three phases of testing and list two errors in each.
- [ ] Practise correct labelling technique with a supervisor.
- [ ] Demonstrate hand hygiene and PPE use in the laboratory.
- [ ] Interpret a simple Levey-Jennings chart and state when to stop testing.
- [ ] Learn ten laboratory safety rules and explain the reason for each.
- [ ] Complete one laboratory practical report with methods, observations and sources of error.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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